LAB EXPERIMENT RECORD

13. Dataset: Geographic Data

Department	AIDS
Course Code / Course Name	DSA0602/ Data Handling and Visualization
Experiment	Data Visualization using R
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Register Number	192424049
Date	06/10/2026
Day	Wednesday

AIM

To analyze and visualize geographic data using R programming by creating a map chart, scatter plot, and data table, and to prepare an interactive Tableau dashboard for exploring the relationship between population, average temperature, and elevation.

1. Create a map chart to visualize the distribution of cities on a geographic map. Label the map elements..

```
install.packages("maps")
library(ggplot2)
library(maps)

cities <- data.frame(
  City = c("City A", "City B", "City C"),
  Population = c(500000, 700000, 600000),
  Temperature = c(75, 68, 80),
  Longitude = c(-100, -80, -60),
  Latitude = c(20, 35, 45)
)

world_map <- map_data("world")

ggplot() +
  geom_polygon(
    data = world_map,
    aes(x = long, y = lat, group = group),
    fill = "lightgray",
    color = "white"
  ) +
  geom_point(
    data = cities,
    aes(x = Longitude, y = Latitude,
        size = Population, color = Temperature)
  ) +
  geom_text(
    data = cities,
    aes(x = Longitude, y = Latitude, label = City),
    nudge_y = 3
  ) +
  coord_quickmap() +
  labs(
    title = "Geographic Distribution of Cities",
    x = "Longitude",
    y = "Latitude",
    size = "Population",
    color = "Average Temperature"
  ) +
  theme_minimal()
```

```
install.packages("maps")
library(ggplot2)
library(maps)

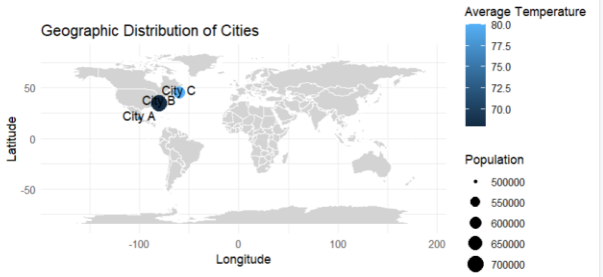
cities <- data.frame(
  City = c("City A", "City B", "City C"),
  Population = c(500000, 700000, 600000),
  Temperature = c(75, 68, 80),
  Longitude = c(-100, -80, -60),
  Latitude = c(20, 35, 45)
)

world_map <- map_data("world")

ggplot() +
  geom_polygon(
    data = world_map,
    aes(x = long, y = lat, group = group),
    fill = "lightgray",
```

```
+   geom_point(
+     data = cities,
+     aes(x = Longitude, y = Latitude,
+         size = Population, color = Temperature)
+   ) +
+   geom_text(
+     data = cities,
+     aes(x = Longitude, y = Latitude, label = City),
+     nudge_y = 3
+   ) +
+   coord_quickmap() +
+   labs(
+     title = "Geographic Distribution of Cities",
+     x = "Longitude",
+     y = "Latitude",
+     size = "Population",
+     color = "Average Temperature"
+   ) +
+   theme_minimal()
>
```

traffic	3 obs. of 2 variables
traffic_data	3 obs. of 3 variables
ts_data	24 obs. of 4 variables
ui	List of 4
weather	5 obs. of 4 variables
website_data	5 obs. of 3 variables
weight_height_data	5 obs. of 2 variables
Wind_mat	num [1:30, 1:30] 12 12 12 12 12 12 12 12 12 12 ...
world_map	99338 obs. of 6 variables

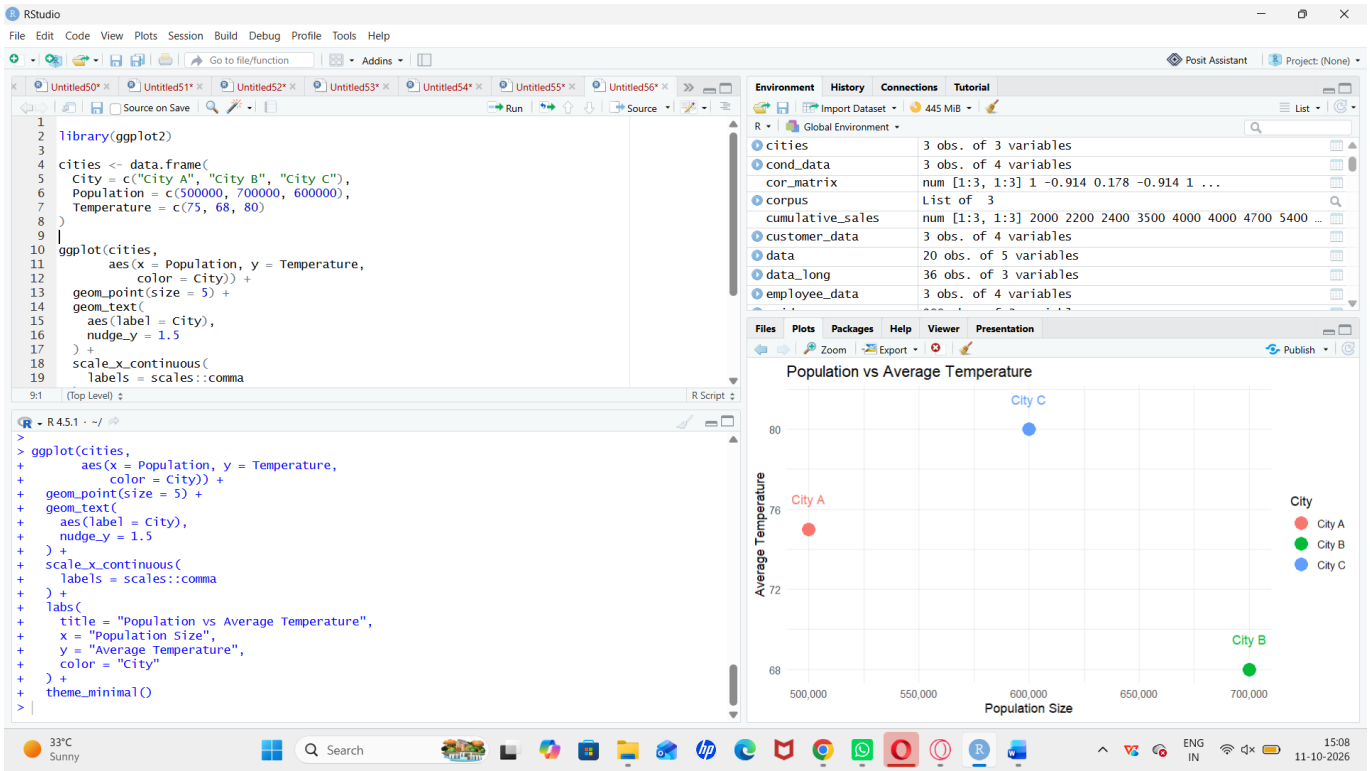


2. Generate a scatter plot to explore the relationship between average temperature and population size.

```
library(ggplot2)

cities <- data.frame(
  City = c("City A", "City B", "City C"),
  Population = c(500000, 700000, 600000),
  Temperature = c(75, 68, 80)
)

ggplot(cities,
       aes(x = Population, y = Temperature,
           color = City)) +
  geom_point(size = 5) +
  geom_text(
    aes(label = City),
    nudge_y = 1.5
  ) +
  scale_x_continuous(
    labels = scales::comma
  ) +
  labs(
    title = "Population vs Average Temperature",
    x = "Population Size",
    y = "Average Temperature",
    color = "City"
  ) +
  theme_minimal()
```

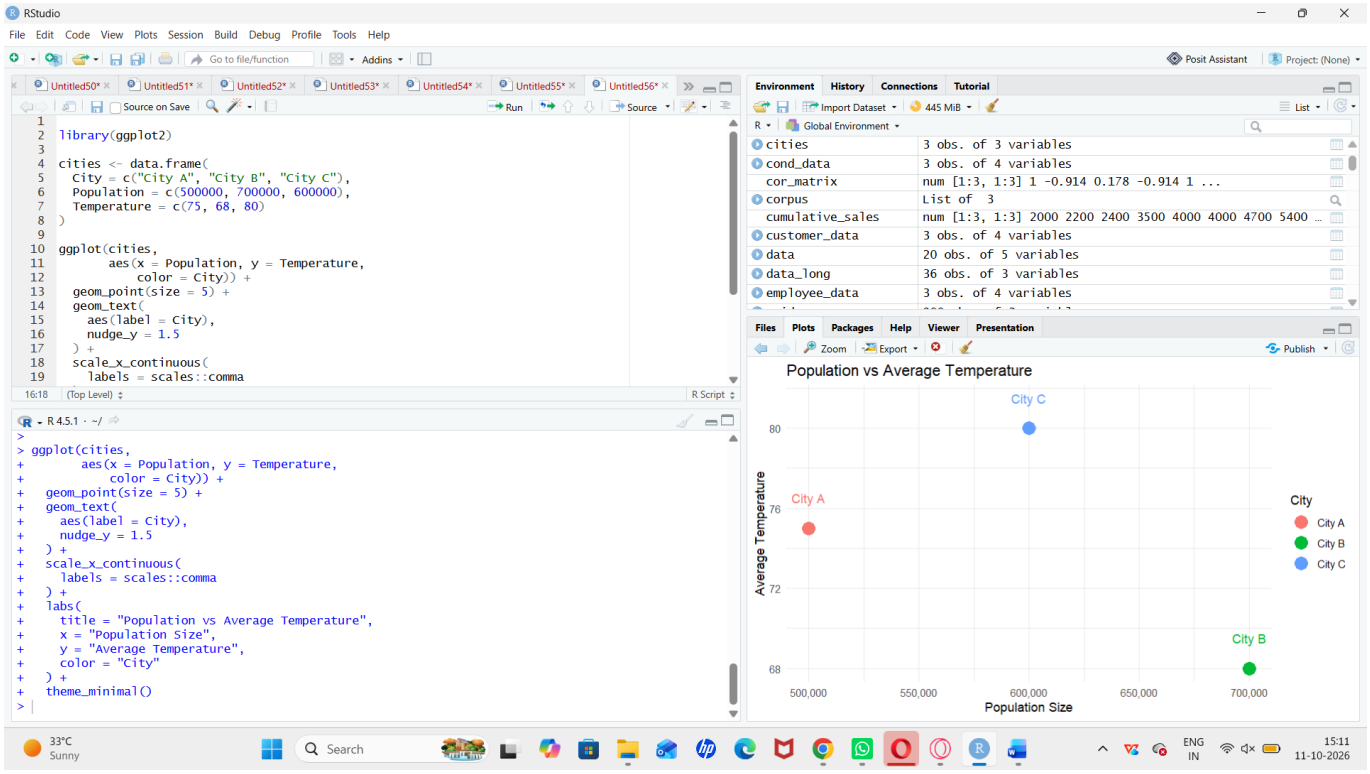


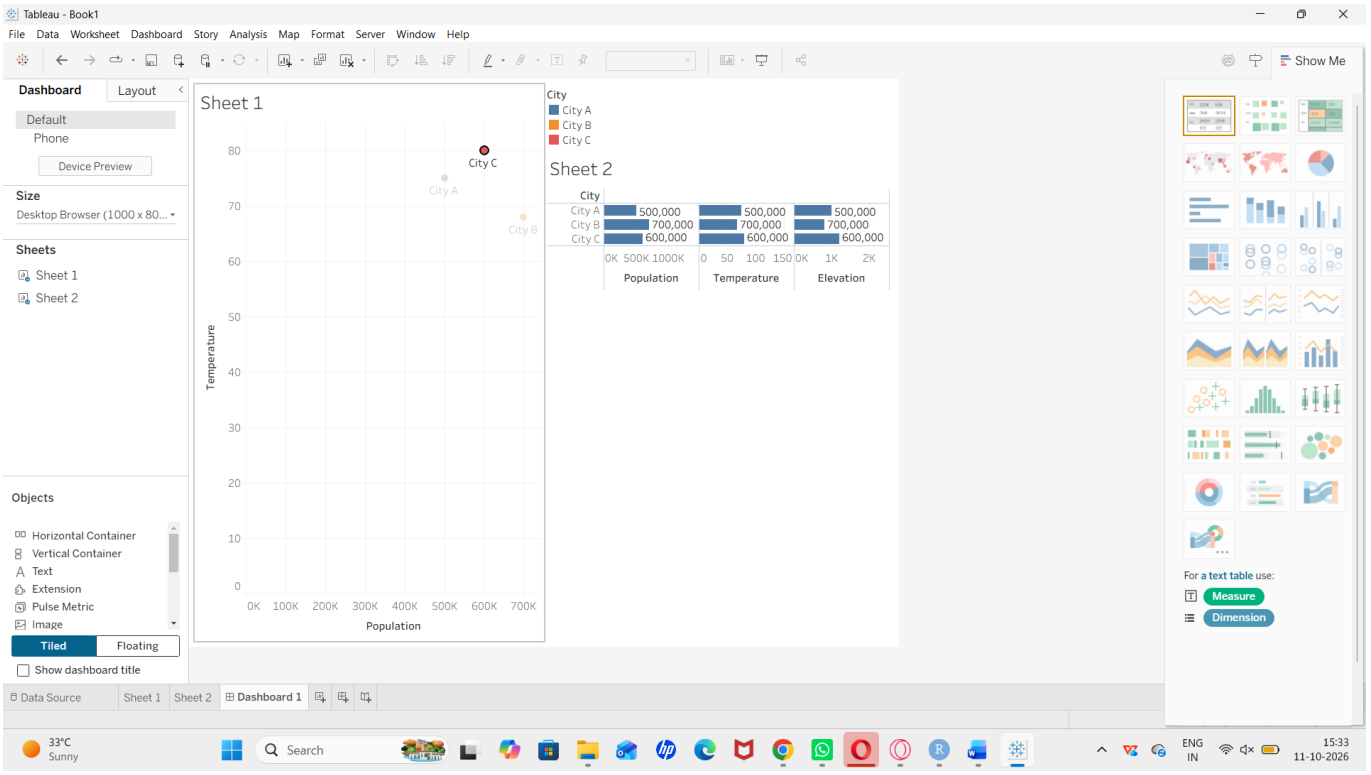
3. Build a table to display the geographic data for each city. Label the table elements.

```
library(ggplot2)

cities <- data.frame(
  City = c("City A", "City B", "City C"),
  Population = c(500000, 700000, 600000),
  Temperature = c(75, 68, 80)
)

ggplot(cities,
       aes(x = Population, y = Temperature,
           color = City)) +
  geom_point(size = 5) +
  geom_text(
    aes(label = City),
    nudge_y = 1.5
  ) +
  scale_x_continuous(
    labels = scales::comma
  ) +
  labs(
    title = "Population vs Average Temperature",
    x = "Population Size",
    y = "Average Temperature",
    color = "City"
  ) +
  theme_minimal()
```





PROCEDURE

1. Create a dataset in R containing city name, population, average temperature, and elevation.

2. Load the required R packages, such as ggplot2, maps, and gridExtra.

3. Create a map chart to display city locations with labels, population-based point sizes, and temperature-based colors.

4. Generate a scatter plot to analyze the relationship between population size and average temperature.

5. Create a table to display the geographic data of all cities.

6. Export the dataset as a CSV file using the write.csv() function.

7. Import the CSV file into Tableau and create separate worksheets for the map, scatter plot, and table.

8. Combine the worksheets into a single dashboard and add filters for interactive data exploration.

RESULT

Thus, geographic data visualization was performed using R by creating a map chart, scatter plot, and data table. The dataset was exported to CSV format and prepared for integration into a Tableau dashboard, enabling interactive exploration and comparison of population, average temperature, and elevation across the cities.